QXU4002 – Chemistry for Materials

Question Set 9

Comments: some questions in the question set are not like exam or test questions. They are used for the purpose of reflecting on some topic to ensure students understand the main concepts. These will be called here “Reflecting questions” (RQ). The questions more similar to exam/test questions will be labelled ETQ.

In class questions:

1) Propose a strategy to incorporate free amino groups (-NH_2) on the surface of crosslinked SU-8
   a) Draw reaction scheme(s) and chemical structure(s) of reagent(s)
   b) Draw the mechanism of the reaction(s) you proposed
   c) Draw the mechanism of the crosslinking of SU-8 using a photoacid under UV-exposure

The final crosslinked SU-8 will still have unreacted epoxide groups on the surface. If you look at LN2, we learnt that epoxides react with amines. If you use an excess of a compound having TWO –NH2 groups, one will react with the epoxy group, but if there is a LARGE excess of –NH2 groups to epoxy groups, the second NH2 should be left unreacted.

2) Propose a strategy to make crosslinked SU-8 as hydrophilic as possible but without reactive functional groups
   a) Draw reaction scheme(s) and chemical structure(s) of reagent(s)
   b) Draw the mechanism of the reaction(s) you proposed

MANY possibilities (look in LN4 in the section about surfactants to see what can be a highly hydrophilic group without reactive groups: you will learn about PEG. SO, you can react the excess of epoxy groups on the surface of the SU-8 with a PEG that has an NH2 at the end. After the reaction, you can react the OH and NH formed with either acetic anhydride (formation of ester and amide: considered unreactive at low temperature) or CH3I to form ethers and tertiary amines/quaternary ammonium (considered unreactive).

Mechanisms in videos and LNs.

3) Propose a strategy to make crosslinked SU-8 as hydrophobic as possible
   a) Draw reaction scheme(s) and chemical structure(s) of reagent(s)
   b) Draw the mechanism of the reaction(s) you proposed

   Similar to previous one, BUT, you want to use an amine with a very long alkyl chain. AFTERWARDS, you want to react the OH and the NH with an acid chloride with a very long alkyl chain (to form ester and amides). This way, there is the most hydrophobic groups possible and no free OH or NH

4) Propose a strategy to make MMA-based photoresist (see Lecture notes for structure) as hydrophobic as possible and incorporate a functional group suitable to react with thiols

   a) Draw reaction scheme(s) and chemical structure(s) of reagent(s)
   b) Draw the mechanism of the reaction(s) you proposed
   Use an acid to deprotect the group, generating COOH. This is very hydrophilic, BUT, if you make an acid chloride with it, AND then make an ester with an alcohol with a VERY long alkyl chain: you have no free reactive groups and the system will be very hydrophobic.